



Data Modelling



Today we'll cover

- **What is Data Modelling**
- **Data Modelling Process**
- **Stages of Data Modelling**
- **How to create an Entities Relationship Model**



What is data modelling

So far we haven't really thought about how we can organise the data we're storing, or even had something close to plan when it came to this.

We just created some models, generated some mock data and tested it out, hoping for the best...

Q. Imagine now that you were tasked with building an app, and you're given a list of requirements. How would you go about planning its data implementation?



What is data modelling

This is when data modelling comes into play!

The **goal** is to **illustrate** the types of **data** used and stored within our system, the **relationships** among these data types, the ways the data can be **grouped** and **organized** and its formats and attributes.



What is data modelling

Data modeling makes it **easier** for developers, data architects, business analysts, and other stakeholders to **view** and **understand** relationships among the **data** in a database.

In addition, it can:

- Reduce errors in software and database development.
- Increase consistency in documentation and system design across the enterprise.
- Improve application and database performance.
- Ease data mapping throughout the organization.
- Improve communication between developers and business intelligence teams.



What is data modelling

Data models are built around **business needs**!

Rules and **requirements** are defined upfront through feedback from **business stakeholders** so they can be incorporated into the **design** of a new system or adapted in the iteration of an existing one.



What is data modelling

The process **begins** by collecting information about **business requirements** from **stakeholders** and **end users**.

These business rules are then **translated** into **data structures** to formulate a concrete database design.

A data model can be **compared** to a roadmap, an architect's blueprint or any formal **diagram** that facilitates a deeper understanding of what is being designed.



Data Modelling Process

- **Identify the entities.** The process of data modeling begins with the identification of the things, events or concepts that are represented in the data set that is to be modeled. Each entity should be cohesive and logically discrete from all others.
- **Identify key properties of each entity.** Each entity type can be differentiated from all others because it has one or more unique properties, called attributes.



Data Modelling Process

- **Identify relationships among entities.** The earliest draft of a data model will specify the nature of the relationships each entity has with the others.
- **Map attributes to entities completely.** This will ensure the model reflects how the business will use the data.



Data Modelling Process

- **Assign keys as needed, and decide on a degree of normalization.** Normalization is a technique for organizing data models (and the databases they represent) in which numerical identifiers, called keys, are assigned to groups of data to represent relationships between them without repeating the data.



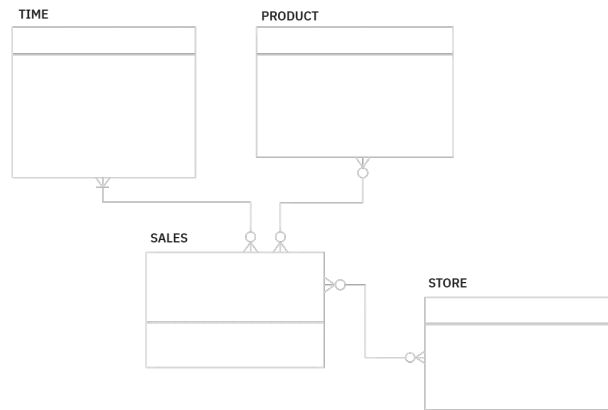
Stages of Data Modelling

We start with some **Conceptual data models**.

They offer a **big-picture** of what the system will contain, how it will be organized, and which business rules are involved.

Any **notation** is typically **simple** at this stage.

CONCEPTUAL DATA MODELING





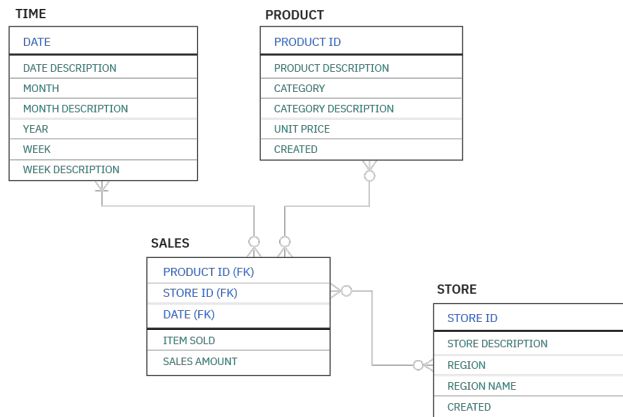
Stages of Data Modelling

After we move into **Logical data models**. They provide greater detail about the concepts and relationships in the domain under consideration.

These indicate data attributes, such as data types and their corresponding lengths, and show the relationships among entities.

Logical data models don't specify any technical system requirements.

LOGICAL DATA MODELING



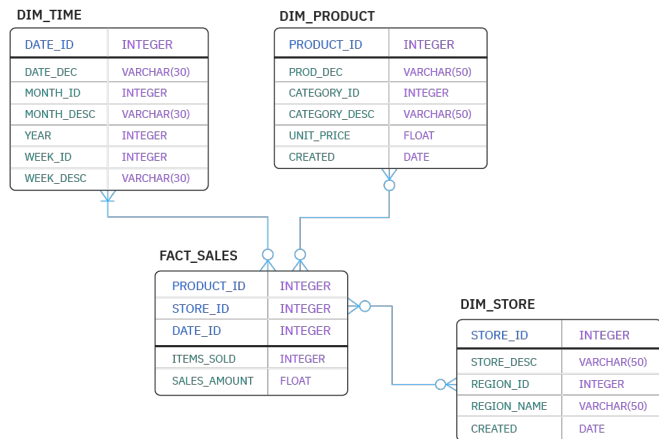


Stages of Data Modelling

Then we finalise with **Physical data models**.

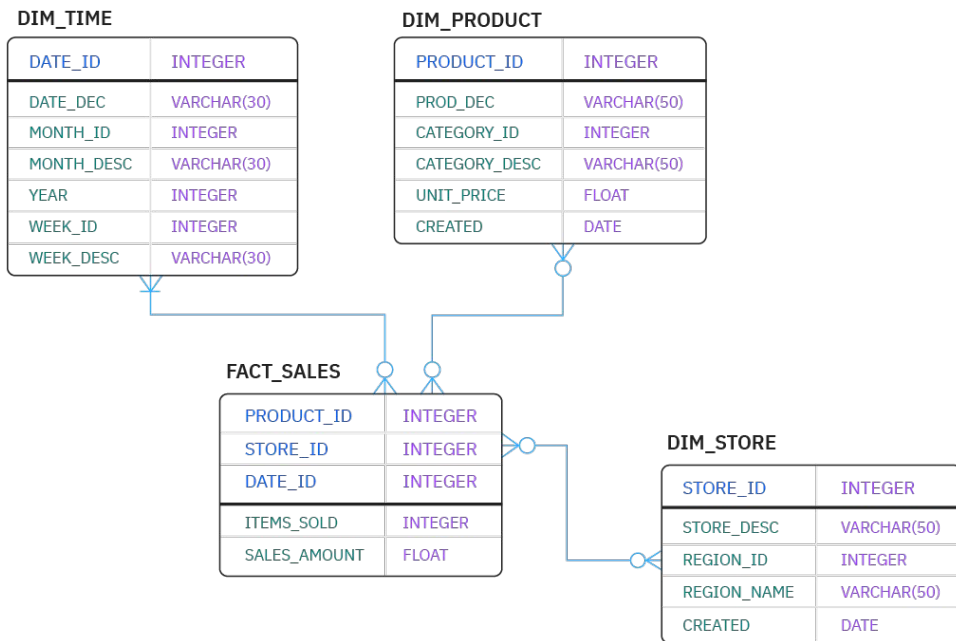
They offer a finalized design that can be implemented as a database, including associative tables that illustrate the relationships among entities as well as the primary keys and foreign keys that will be used to maintain those relationships.

PHYSICAL DATA MODELING





PHYSICAL DATA MODELING





How to create an Entities Relationship Model

Entity-relationship (ER) data models use formal diagrams to represent the relationships between entities in a database. Several ER modeling tools are used by data architects to create visual maps that convey database design objectives.

We're going to use draw.io to help us build our ER model!



Time for some live-coding

A local hotel wants to keep a record of room bookings. The hotel has a lot of rooms, but customers will only be able to choose one room per booking. The person who books the room is not necessarily the person providing contact details.

Lets create an Entity Relationship Diagram !



Time for some practice

Let's create some ER Models for:

- **Twitter**
- **Amazon**
- **Github**